

THE SEVENTEENTH WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

March 2, 1957

Morning Session

1. The normals to a surface all intersect a fixed straight line. Show that the surface is a portion of a surface of revolution. (page 433)

2. A uniform wire is bent into a form coinciding with the portion of the curve $y = e^x$, $0 \leq x \leq a$, $a > 1$, and the line segment $a - 1 \leq x \leq a$, $y = e^a$. The wire is then suspended from the point $(a - 1, e^a)$ and a horizontal force F is applied at the point $(0, 1)$ to hold the wire in coincidence with the curve and segment. Assuming the x axis is horizontal, show that the force F is directed to the right. (page 436)

3. A and B are real numbers and k a positive integer. Show that

$$\left| \frac{\cos kB \cos A - \cos kA \cos B}{\cos B - \cos A} \right| < k^2 - 1$$

whenever the left side is defined. (page 438)

4. $P(z)$ is a complex polynomial whose roots (as points in the Argand plane) can be covered by a closed circular disc of radius R . Show that the roots of $nP(z) - kP'(z)$ can be covered by a closed circular disc of radius $R + |k|$, where n is the degree of $P(z)$, k is any complex number, and $P'(z)$ is the derivative of $P(z)$. (page 439)

5. Given n points in the plane, show that the largest distance determined by these points cannot occur more than n times. (page 440)

6. $S_1 = \ln a$ and $S_n = \sum_{i=1}^{n-1} \ln (a - S_i)$, $n > 1$.

Show that

$$\lim_{n \rightarrow \infty} S_n = a - 1. \quad (\text{page 441})$$

7. Each member of a set of circles in the xy plane is tangent to the x axis and no two of the circles intersect. Show that:

(i) the points of tangency can include all the rational points on the axis, but

(ii) the points of tangency cannot include all the irrational points.

(page 443)

Afternoon Session

1. Consider the determinant $|a_{ij}|$ of order 100 with $a_{ij} = i \times j$. Prove that if the absolute value of each of the $100!$ terms in the expansion of this determinant is divided by 101 then the remainder in each case is 1. (page 444)

2. If facilities for division are not available, it is sometimes convenient in determining the decimal expansion of $1/A$, $A > 0$ to use the iteration $X_{k+1} = X_k(2 - AX_k)$, $k = 0, 1, 2, \dots$, where X_0 is a selected "starting" value. Find the limitations, if any, on the starting value X_0 in order that the above iteration converges to the desired value $1/A$. (page 444)

3. For $f(x)$ a positive, monotone decreasing function defined in $0 \leq x \leq 1$ prove that

$$\frac{\int_0^1 xf^2(x) dx}{\int_0^1 xf(x) dx} \leq \frac{\int_0^1 f^2(x) dx}{\int_0^1 f(x) dx}. \quad (\text{page 446})$$

4. Let $a(n)$ be the number of representations of the positive integer n as the sums of 1's and 2's taking order into account. For example, since

$$\begin{aligned} 4 &= 1 + 1 + 2 = 1 + 2 + 1 = 2 + 1 + 1 \\ &= 2 + 2 = 1 + 1 + 1 + 1, \end{aligned}$$

then $a(4) = 5$. Let $b(n)$ be the number of representations of n as the sum of integers greater than 1, again taking order into account and counting the summand n . For example, since $6 = 4 + 2 = 2 + 4 = 3 + 3 = 2 + 2 + 2$, we have $b(6) = 5$. Show that for each n , $a(n) = b(n + 2)$. (page 447)

5. With each subset X of a set is associated a second subset $f(X)$. The association is such that whenever X contains Y then $f(X)$ contains $f(Y)$. Show that for some set A , $f(A) = A$. (page 449)

6. The curve $y = f(x)$ passes through the origin with a slope of 1. It satisfies the differential equation $(x^2 + 9)y'' + (x^2 + 4)y = 0$. Show that it crosses the x axis between

$$x = \frac{3}{2} \pi \quad \text{and} \quad x = \sqrt{\frac{63}{53}} \pi. \quad (\text{page 450})$$

7. Let C be a closed convex planar disc bounded by a regular polygon. Show that for each positive integer n there exists a set of points $S(n)$ in the plane such that each n points of $S(n)$ can be covered by C , but $S(n)$ itself cannot be covered by C . (page 453)